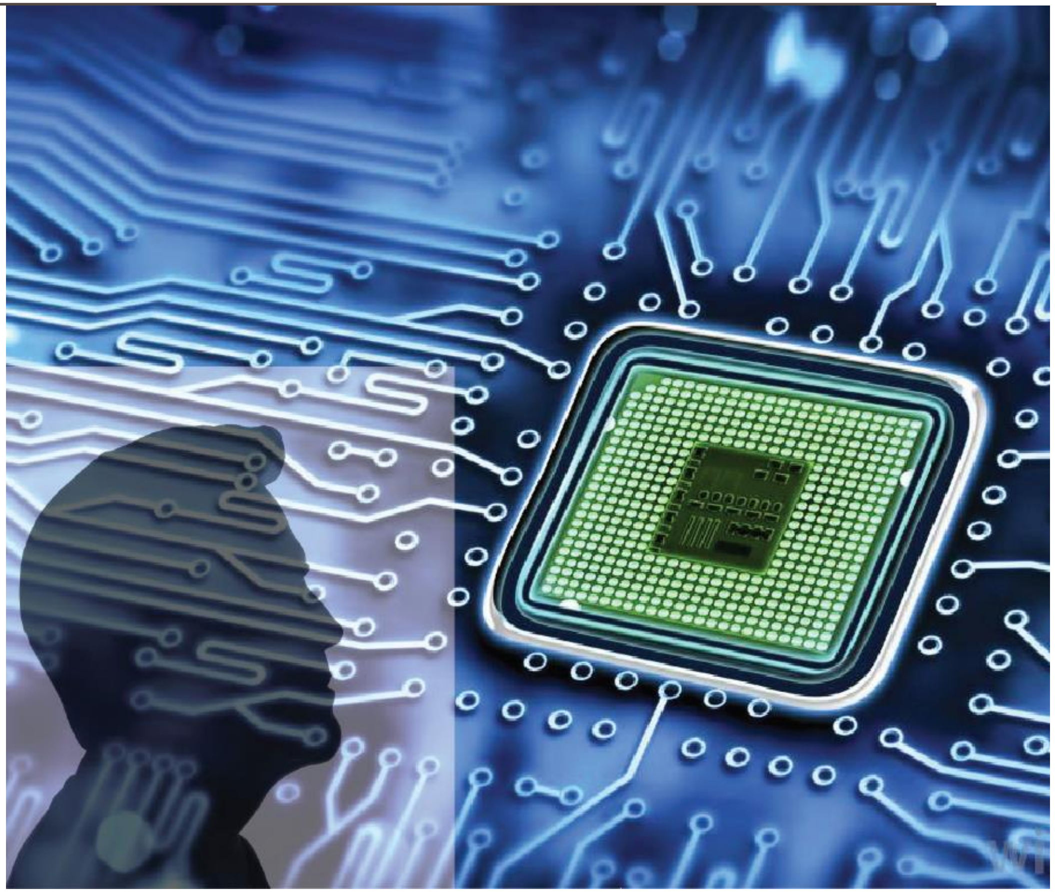
VLSI DESIGN FLOW

Week-12 Lecture-1
Physical Design



Sneh Saurabh
11th November, 2024

VDF: Revision/Announcements

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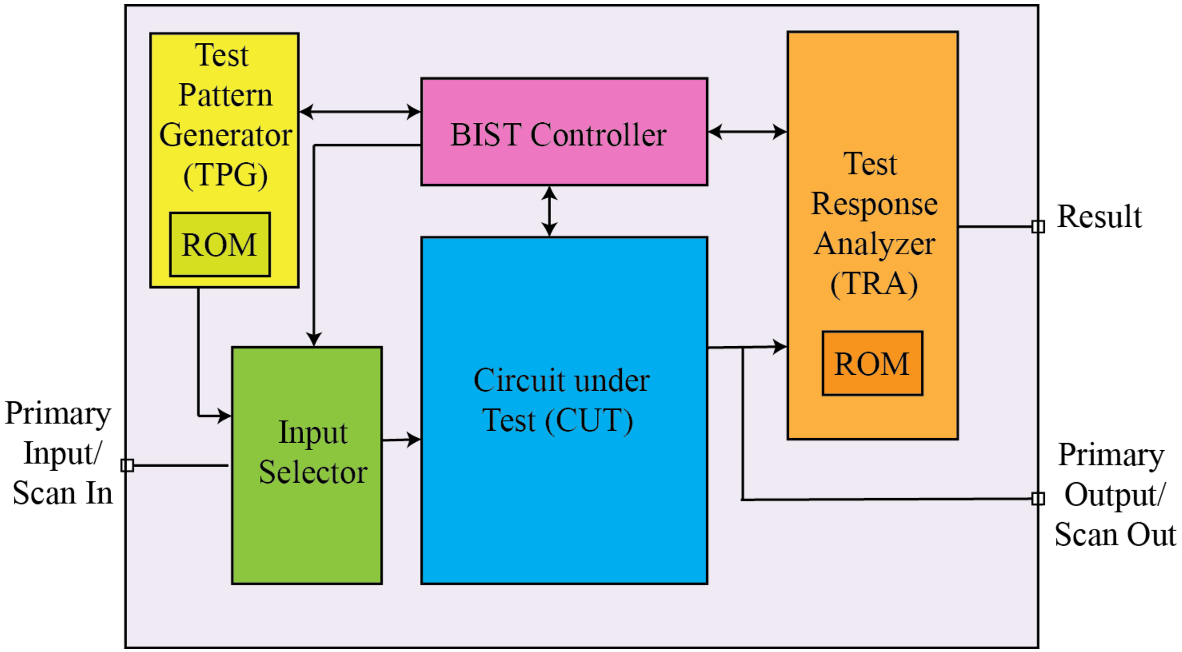
Design for Test

BIST

BIST: Architecture



- **Test Pattern Generator (TPG)**
  - Generates pattern for testing
  - Can contain ROM
- **Test Response Analyzer (TRA)**
  - Generates signature for the test response and compare
  - ROM stores golden signature
  - Compares test signature with the golden signature
- **Test Controller**
  - Controls all the operations of BIST
  - Strategy for overall test control is most difficult part of BIST

BIST: Test Response Analyzer (1)

- No. of test patterns can be large (e.g. 1 million) and there can be many outputs (e.g. 100)
  - Too **voluminous** data to store in ROM (100 million bits)

- **Compaction**: method of reducing the number of bits in the circuit response

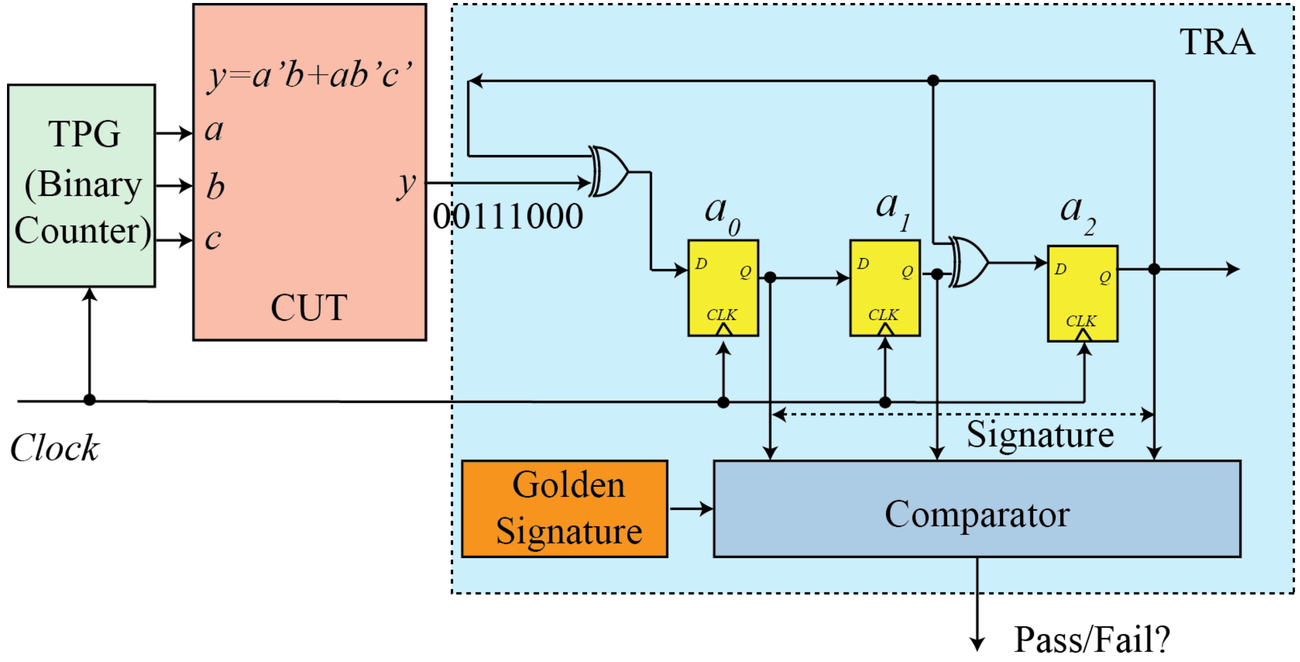
- **Signature**: a statistical property of a circuit, usually a number computed for a circuit from its responses

- **Signature Analysis**: signature of a good circuit ("golden signature") compared with the signature of a "potentially" faulty circuit ("test signature")
  - If "golden signature" matches "test signature" then the circuit is assumed to be good, else faulty
  - Desirable: signature of a good circuit and faulty circuits are different

- **Aliasing**: During testing, it is possible that the signature of "good" circuit and "faulty" circuit match
  - Due to information loss during compaction

BIST: Test Response Analyzer

- Techniques to compute Signature
  - **Ones Counting**: count number of ones across all circuit responses
    - Different permutations can yield same signature
  - **Modular LFSR**: Extra XOR gate at the input
    - Single Input Signature Register (SISR)
    - Need to start with a *seed value*



Built-in Self Test (BIST)

Advantages

- Lower cost of test, since the need for external electrical testing using an ATE will be reduced
- No need to store test patterns
- At-speed test can be done
- Capability to perform tests outside the production electrical testing environment

Disadvantages

- Additional silicon area
  - Yield and Reliability decrease
- Performance loss
  - Extra delays
- Additional pin (and possibly bigger package size) requirements

VDF: Course Content

Week No.	Topic
1	Introduction: Basic Concepts; Design Styles; Designing vs. Fabrication; Processes involved in taking "Idea" to RTL: Software/Hardware Partitioning, Behavioral Synthesis
2	Overview of "RTL to GDS" Design, Verification and Test Flow; Processes after GDS Tapeout to Final Chip;
3	Design capture: HDL (Verilog), Library and Constraints;
4	RTL Simulation and Synthesis; **Static Timing Analysis; Power Analysis;**
5	**Introduction to Formal Methods: Equivalence Checking, BDD, SAT;**
6	Logic Synthesis: Optimization, Technology Mapping;
7	Design for Test (DFT): Fault Models, Scan Insertion, ATPG, BIST
8	**Overview of CMOS Fabrication Technology**; Basics of Physical Design
9	Floorplanning, Power Planning;
10	Placement: Techniques and Optimization;
11	Clock Tree Synthesis: Distribution, Skew Optimization; Routing: Global and Detailed; ECO
12	**Physical Verification: Extraction, LVS, ERC, DRC; Sign-off**
13	Summing-up and Paper Presentations

Basic Concepts for Physical Design

Physical Concepts for Physical Design

IC Fabrication

IC Fabrication

- Photolithography is the key step in fabrication

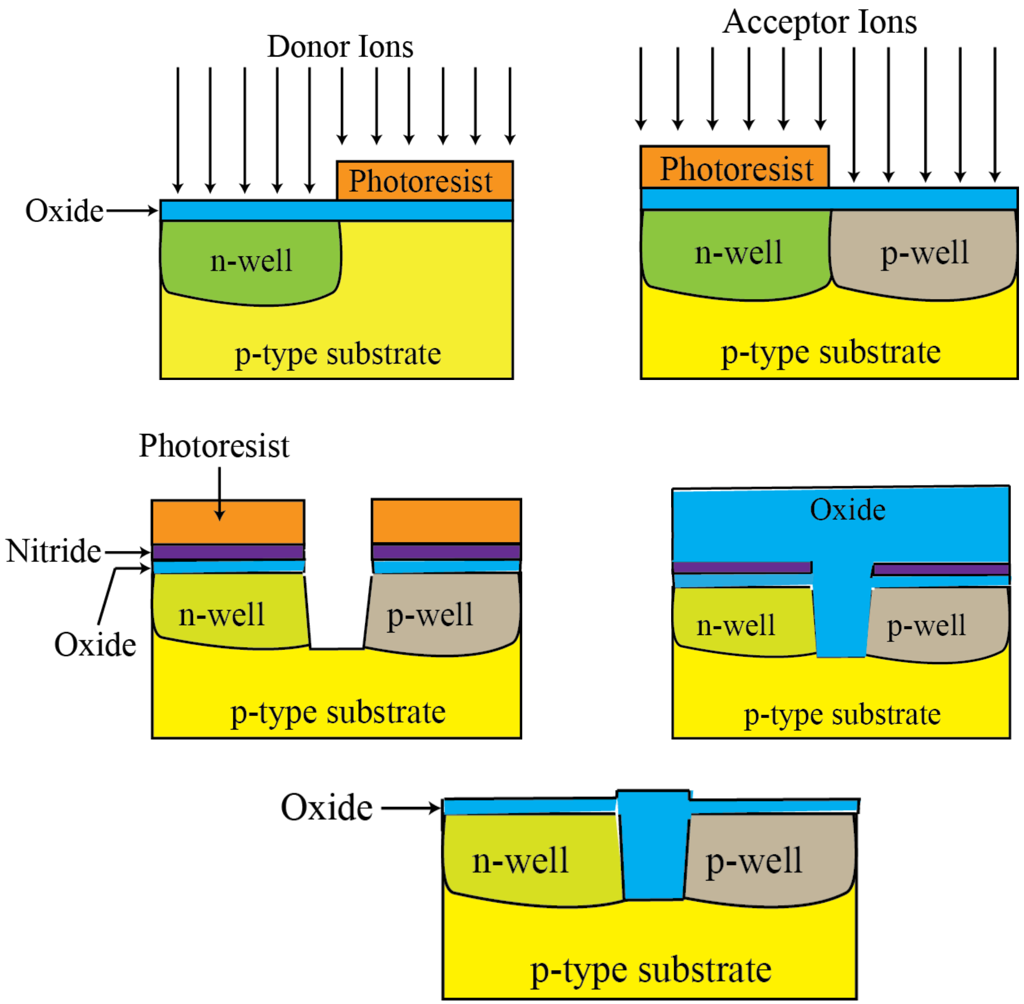
Two types of process:

1. Front End Of the Line (FEOL): active elements (transistors, diodes, capacitors)
2. Back End Of the Line (BEOL): layers of wires

FEOL Processes (1)

- Doping with acceptor and donor atoms
- Ion Implantation:
  - ➢ Ion created, using arc discharge and accelerated/filtered using electric and magnetic fields
  - ➢ bombard the substrate through a thin layer of screening silicon dioxide
- Annealing: activate dopants
- Threshold voltage adjustment implants

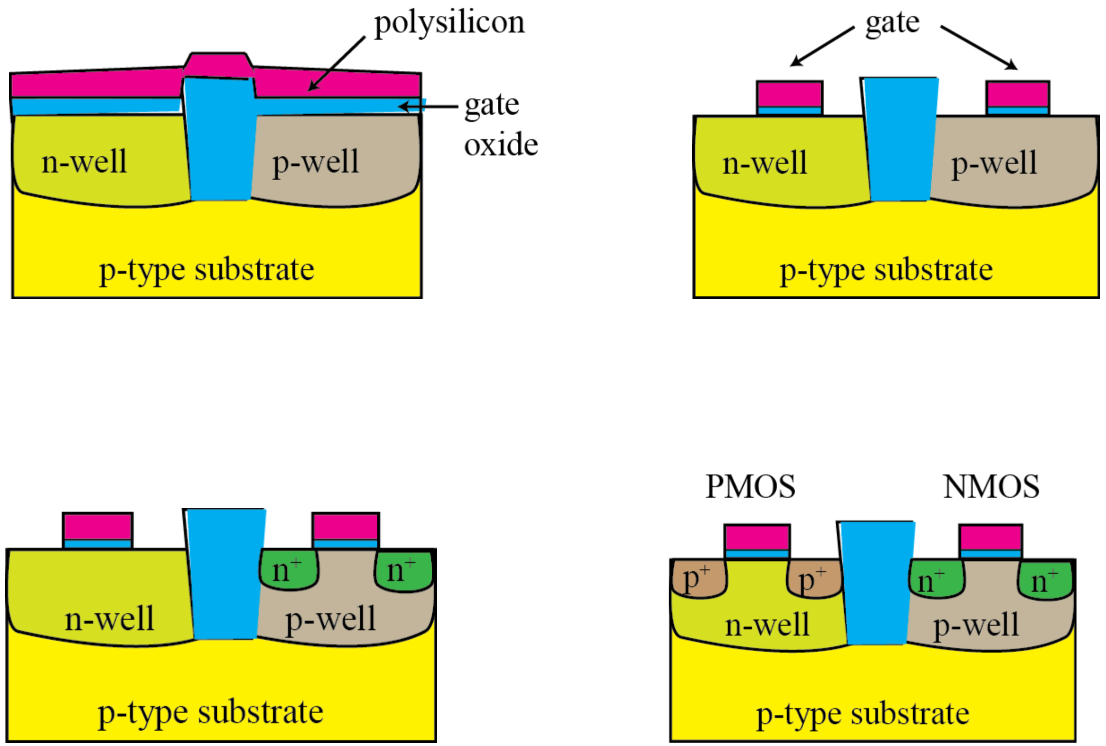
- **Isolation**: ensures that the transistors do not interact through the substrate
- Shallow Trench Isolation (STI)



FEOL Processes (2)

Transistor formation (gate, source and drain)

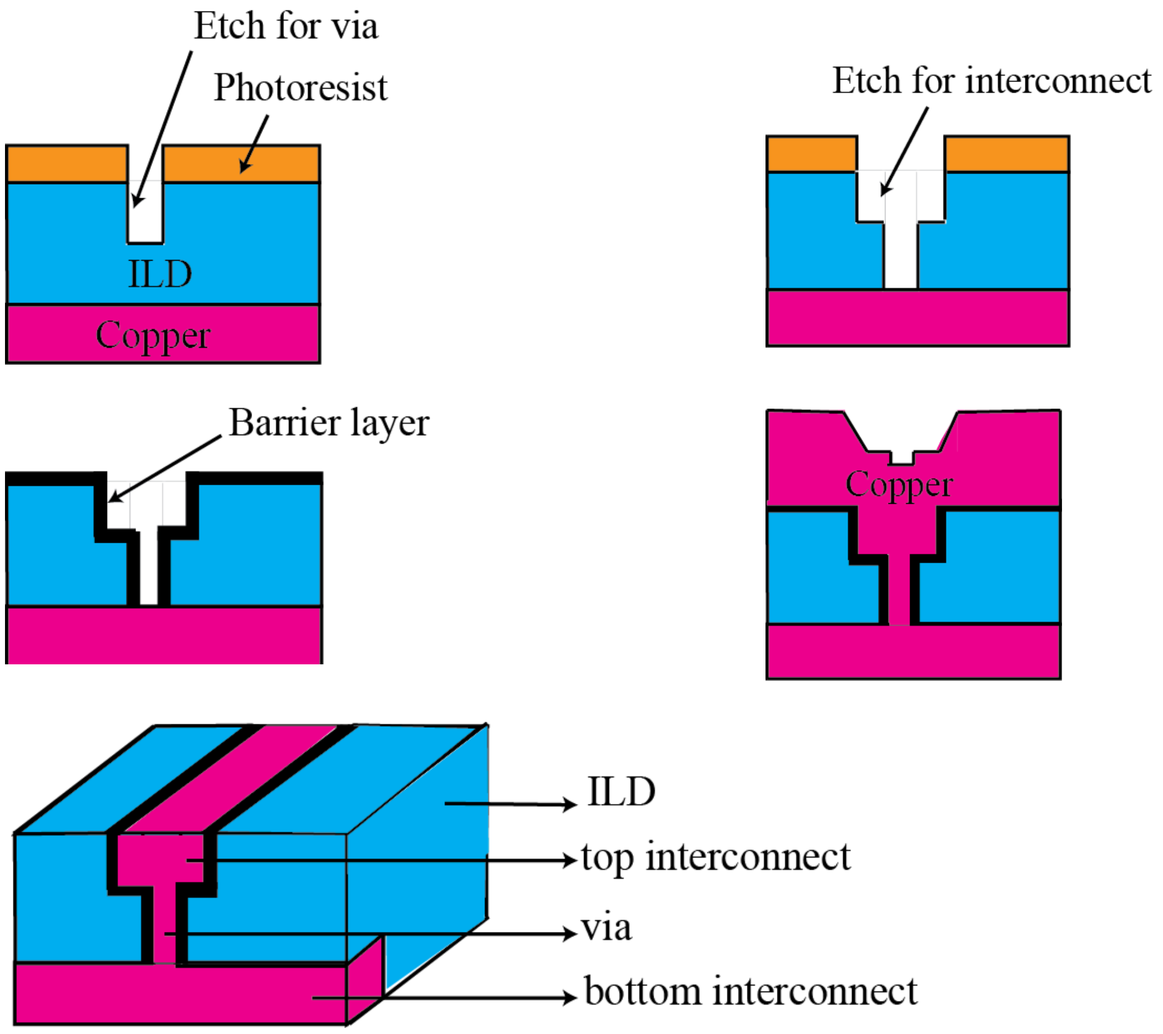
- High quality gate oxide growth/deposition over silicon channel
- Polysilicon deposition for gate contact
  - Chemical vapor deposition (CVD)
- Patterning
  - Local interconnect
- Source-drain Ion Implantation and annealing for NMOS and PMOS



BEOL Processes (1)

Back End Of the Line (BEOL):

- Layers of Wires
- Inter Layer Dielectric (ILD)
- Copper
  - ➢ Dual Damascene Process
  - ➢ Repeated for each layer



References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.
- M. Bushnell and V. Agrawal, “Essentials of electronic testing for digital, memory and mixed-signal VLSI circuits”, Springer Science & Business Media, 2004.